Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Final, Spring 2022 — Problem 1

(i) Expand along the first row:

$$\det = x(2\cdot 3 - 1\cdot 0) - 1(3y - z) + 0 = \boxed{6x - 3y + z}.$$

(ii) (x, y, z) lies in P exactly when it is a combination of $(1, 2, 0)$ and $(0, 1, 3)$, i.e. when the three columns are dependent, i.e. when that determinant is 0:

$$\boxed{a = 6,\ b = -3,\ c = 1} \quad (6x - 3y + z = 0;\ \text{check } (1, 2, 0) \text{ and } (0, 1, 3)).$$

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Math 208 Final, Spring 2022 — Problem 2

(a) A is upper triangular, so its eigenvalues are the diagonal entries: $\boxed{\lambda = 2 \text{ (twice)},\ \lambda = 3}$.

(b) $\lambda = 2$: $A - 2I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ gives $x_2 = x_3 = 0$, basis $\boxed{\{(1,0,0)\}}$. $\lambda = 3$: $A - 3I = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ gives $x_1 = x_2 = x_3$, basis $\boxed{\{(1,1,1)\}}$.

(c) **No**: only two independent eigenvectors in $\mathbb{R}^3$.

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Math 208 Final, Spring 2022 — Problem 3

A swap multiplies the determinant by -1, row replacements don't change it, and scaling a row by $\frac{1}{2}$ multiplies it by $\frac{1}{2}$. So $\det B = (-1)(\frac{1}{2})\det A$. B is triangular with $\det B = 1 \cdot 1 \cdot x = x$:

$$\boxed{\det A = -2x}.$$

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Math 208 Final, Spring 2022 — Problem 4

A is 6×6 with eigenvalues 0 (multiplicity 3), 1 (multiplicity 2) and -1 (multiplicity 1).

(a) $\boxed{-1,\ 0,\ 1}$.

(b) The nullity is $\dim E_0$, between 1 and the multiplicity 3: $\boxed{1,\ 2,\ 3}$.

(c) $A^2 - I = (A-I)(A+I)$, and its null space is $E_1 \oplus E_{-1}$, of dimension $\dim E_1 + \dim E_{-1} \in \{1,2\} + 1 = \{2,3\}$. (The eigenvalues of A^2 are $0,0,0,1,1,1$.) So $\operatorname{rank}(A^2 - I) = 6 - \text{nullity} \in \boxed{\{3,\ 4\}}$.

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Math 208 Final, Spring 2022 — Problem 5

(a) $\boxed{A = \begin{bmatrix} 1 & 0 & 4 & -1 \\ 0 & 1 & 1 & 2 \end{bmatrix}}$.

(b) A is already reduced; z, w are free: $x = -4z + w$, $y = -z - 2w$.

$$\ker T = \operatorname{span}\{(-4, -1, 1, 0),\ (1, -2, 0, 1)\}.$$

(c) **No**: the kernel contains nonzero vectors (and a map $\mathbb{R}^4 \to \mathbb{R}^2$ can never be one-to-one).

Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Final, Spring 2022 — Problem 6

(a) $A\begin{bmatrix}2\\2\\2\end{bmatrix} = \begin{bmatrix}2x+2y\\2x+2\\2y+2\end{bmatrix} = \lambda\begin{bmatrix}2\\2\\2\end{bmatrix}$ means $x+y = x+1 = y+1 = \lambda$. So $x = y = 1$ (with $\lambda = 2$): $\boxed{x = 1,\ y = 1}$.

(b) $\det(A^2) = (\det A)^2 = (x^2+y^2)^2 = 0$ exactly when $\boxed{x = y = 0}$.

Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Final, Spring 2022 — Problem 7

(a) **Not possible**: by linearity $T(1,0) = (3,1)$ and $T(1,1) = (2,2)$ force $T(0,1) = (-1,1)$, so $T(2,1) = 2(3,1) + (-1,1) = (5,3) \neq (0,3)$.

(b) **Not possible**: a 2×2 matrix has at most 2 eigenvalues.

(c) For example $\boxed{B = \begin{bmatrix} 8 & 0 & 0 \\ 8 & 0 & 0 \\ 8 & 0 & 0 \end{bmatrix}}$: $B(1,1,1) = (8,8,8)$ and $\det B = 0$.

Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Final, Spring 2022 — Problem 8

(a) The first step is wrong: the determinant is not additive, so $\det(A - \lambda I) \neq \det A - \det(\lambda I)$ in general.

(b) For example $\boxed{A = \begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}}$: $\det A = 1$, eigenvalues 2 and $\frac{1}{2}$.

Math 208 Final, Spring 2022 — Problem 9

(a) $A\begin{bmatrix}3\\1\end{bmatrix} = \begin{bmatrix}9\\3\end{bmatrix} = 3\begin{bmatrix}3\\1\end{bmatrix}$ and $A\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}-1\\1\end{bmatrix} = -1\begin{bmatrix}1\\-1\end{bmatrix}$:

$$\boxed{P = \begin{bmatrix}3 & 1\\1 & -1\end{bmatrix},\quad D = \begin{bmatrix}3 & 0\\0 & -1\end{bmatrix}}.$$

(b) **Not possible**. If $C = QEQ^{-1}$ with E real diagonal, then $A = C^2 = QE^2Q^{-1}$ would have eigenvalues that are squares of real numbers, so ≥ 0. But A has eigenvalue -1.

(c) Solve $a(3,1) + b(1,-1) = (1,0)$: $a = b = \frac{1}{4}$, so $\boxed{[\vec{x}]_{\mathcal{B}} = (\frac{1}{4}, \frac{1}{4})}$.

(d) **No**: a subspace must contain $\vec{0}$, but $\vec{0}$ is not an eigenvector, so $\vec{0} \notin S$.

Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Final, Spring 2022 — Problem 10

(a) $3B = A + I$ has eigenvalues $3 \cdot 0 = 0$ and $3 \cdot 1 = 3$. An eigenvalue μ of $A + I$ corresponds to $\mu - 1$ for A, so A has eigenvalues $\boxed{-1 \text{ and } 2}$ (check: $\det A = 2 = (-1)^2 \cdot 2$).

(b) $A_n = J - I$ where J is the all-ones matrix, whose eigenvalues are n (once, eigenvector $(1, \ldots, 1)$) and 0 ($n - 1$ times). So A_n has eigenvalues $n - 1$ (once) and -1 ($n - 1$ times):

$$\boxed{\det A_n = (-1)^{n-1}(n-1)} \quad (n = 3: \ 2 \checkmark).$$